
Statistical Inference

B. Statistical Data Science 2nd Year Indian Statistical Institute

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Exercise Series 12

Exercise 1. For the following examples, develop the likelihood ratio test, Wald's test and Rao's score test for testing $\mathcal{H}_0 : \theta = \theta_0$ against $\mathcal{H}_{\text{left}} : \theta < \theta_0$, $\mathcal{H}_{\text{right}} : \theta > \theta_0$ and $\mathcal{H}_{\text{both}} : \theta \neq \theta_0$. Clearly specify the test statistic and the cut-off value. Check whether the regularity conditions required for the asymptotic results hold. In each case, draw the power function of the three tests on the same graph for $n = 5, 10, 25, 50, 100$ (fixing some specific value of θ_0). Based on these plots, what can you say about the relative merits/demerits of the three tests? Do your conclusions depend on the particular value of θ_0 ?

- (a) $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} \text{Binomial}(p)$, $0 \leq p \leq 1$. $\theta = p$.
- (b) $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} \text{Poisson}(\lambda)$, $\lambda > 0$. $\theta = \lambda$
- (c) $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} \text{Geometric}(p)$, $0 \leq p \leq 1$. $\theta = p$.
- (d) $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} \text{Uniform}(0, \theta)$, $\theta > 0$.
- (e) $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} \text{Normal}(\mu, 1)$, $\mu \in \mathbb{R}$. $\theta = \mu$.
- (f) $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} \text{Normal}(0, \sigma^2)$, $\sigma > 0$.
- (g) $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} \text{Exponential}(\theta)$, $\theta > 0$.
- (h) $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} \text{Laplace}(\mu, 1)$, $\mu \in \mathbb{R}$. $\theta = \mu$.
- (i) $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} \text{Laplace}(0, \sigma)$, $\sigma > 0$. $\theta = \sigma$.
- (j) $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} \text{Normal}(\theta, \theta^2)$, $\theta \in \mathbb{R}$.
- (k) $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} \text{Normal}(\theta, \theta)$, $\theta > 0$.

Exercise 2. Consider the linear regression model $Y_i \stackrel{\text{indep.}}{\sim} \text{Normal}(\alpha + \beta x_i, \sigma^2)$, $i = 1, \dots, n$, where $\alpha, \beta \in \mathbb{R}$ and $\sigma^2 > 0$ are unknown parameters. We are interested in testing $\mathcal{H}_0 : \beta = 0$ vs. $\mathcal{H}_a : \beta \neq 0$.

- (a) Formulate the likelihood ratio test, Wald's test and Rao's score test for this problem.
- (b) Compare the power of these three tests numerically.
- (c) Plot the power functions of the tests in part (a) when $Y_i \stackrel{\text{indep.}}{\sim} \text{Laplace}(\alpha + \beta x_i, \sigma)$, $i = 1, \dots, n$. What do you observe?
- (d) Repeat part (c) with $Y_i \stackrel{\text{indep.}}{\sim} \text{Cauchy}(\alpha + \beta x_i, \sigma)$, $i = 1, \dots, n$.

Note: The goal of parts (c) and (d) is to study the effect of misspecification on the inferential procedure. The effect can be severe in many cases.

(e) Repeat parts (a) and (b) for testing $\mathcal{H}_0 : (\alpha, \beta) = (0, 0)$ vs. $\mathcal{H}_a : (\alpha, \beta) \neq (0, 0)$.

Exercise 3. Recall the table giving the number of persons (out of a total of 500) having four different blood groups A, B, AB and O, which we considered in Series 2, Exercise 5 and Series 7, Exercise 6. Such tables are known as 2×2 *contingency tables* and are generally of the following form.

	$B = b_1$	$B = b_2$	Total
$A = a_1$	n_{11}	n_{12}	$n_{1\cdot}$
$A = a_2$	n_{21}	n_{22}	$n_{2\cdot}$
Total	$n_{\cdot 1}$	$n_{\cdot 2}$	n

Here, we have two attributes A and B , each having two levels a_1, a_2 and b_1, b_2 , respectively. In a collection of n items, n_{ij} denotes the number of items having attributes a_i and b_j , $i, j = 1, 2$. The number $n_{i\cdot} = n_{i1} + n_{i2}$ denotes the total number of items having attribute a_i , $i = 1, 2$. Similarly, $n_{\cdot j} = n_{1j} + n_{2j}$ denotes the total number of items having attribute b_j , $j = 1, 2$. It also holds that $n_{11} + n_{12} + n_{21} + n_{22} = n_{1\cdot} + n_{2\cdot} = n_{\cdot 1} + n_{\cdot 2} = n$. If we denote p_{ij} to be the probability of a randomly selected item having attributes a_i and b_j , and assume that the attributes are assigned independently and identically among the items, then we can use a **Multinomial** distribution to model the data. We already did this in Series 7, Exercise 6.

Here, we are interested in checking whether the attributes A and B are independent. That is, the level of the attribute A (a_1 or a_2) in an item does not affect the level of the attribute B (b_1 or b_2) in the item, and vice versa. Defining $p_{i\cdot} = p_{i1} + p_{i2}$ and $p_{\cdot j} = p_{1j} + p_{2j}$, $i, j = 1, 2$, to be the marginal probabilities, this is same as testing $\mathcal{H}_0 : p_{ij} = p_{i\cdot} p_{\cdot j} \forall i, j = 1, 2$ vs. $\mathcal{H}_a : \mathcal{H}_0^c$. Formulate the likelihood ratio test, Wald's test and Rao's score test for this problem. Clearly specify the test statistics and the cut-off values. What are the similarities/dissimilarities among the tests?